Escola d'Enginyeria de Telecomunicació i Aeroespacial de Castelldefels

UNIVERSITAT POLITÈCNICA DE CATALUNYA BARCELONATECH

# Libraries Documentation

Pau Garcia Calderó
EETAC - UPC
paugarcia32@gmail.com

# Table of Contents

1. Introduction ..... 3
2. Comparison Between Libraries ..... 3
    2.1. Unofficial Heltec ..... 3
        2.1.1. Unnoficial Heltec Modules ..... 3
    2.2. Arduino LoRa ..... 3
        2.2.1. Arduino LoRa Modules ..... 3
    2.3. RadioLib ..... 4
        2.3.1. RadioLib Modules ..... 4
    2.4. Comparison table ..... 5
Bibliography ..... 7
Acronyms ..... 8

# 1. Introduction

In this document, I will compare and analyze which LoRa library is the best for this project. As I want to determine all the possible parameters, I will choose the library that allows the user to select and change more parameters.

All this analysis is considering the use of ESP32-S3 microcontrollers, utilizing the Arduino language and its related libraries.

# 2. Comparison Between Libraries

As of today, I have found only three different libraries that allow Arduino to use LoRa, either using embedded LoRa modules or external modules.

## 2.1. Unofficial Heltec

First, there is an `unofficial Heltec` library [1] that tries to implement all the functions related to Heltec Boards, because the official documentation is not intuitive.

This library, as the documentation says, is supposed to use the `SX-1262 LoRa module`, so I don't know if it would work if you try to use this library with any other module.

Also, in the documentation, we can find that this library reuses the `RadioLib` [2] library. As we will see in Section 2.3, the `RadioLib library` allows the use of many more `SX LoRa modules`, so it could allow us to implement other `SX LoRa modules`.

### 2.1.1. Unnoficial Heltec Modules

| Module | Description |
|---|---|
| SX1262 | Semtech SX1262 sub-GHz transceiver |

Table 1: Heltec Modules List

## 2.2. Arduino LoRa

In second place, we have the `Arduino LoRa library` [3]. This library is also an unofficial library maintained by the comunity, which allows the `Semtech SX1276/77/78/79` based boards.

### 2.2.1. Arduino LoRa Modules

| Module | Description |
|---|---|
| SX127x | series LoRa modules (SX1272, SX1273, SX1276, SX1277, SX1278, SX1279) |
| Dragino | Lora Shield |
| HopeRF | RFM95W, RFM96W, and RFM98W |
| Modtronix | inAir4, inAir9, and inAir9B |

Table 2: Arduino LoRa Modules List

## 2.3. RadioLib

In last place, we have the `RadioLib library` [2]. `RadioLib` allows its users to integrate all sorts of different wireless communication modules & protocols into a single system.

RadioLib natively supports Arduino, but can run in non-Arduino environments as well.

### 2.3.1. RadioLib Modules

| Module | Description |
|---|---|
| CC1101 | FSK radio module |
| LLCC68 | LoRa module |
| LR11x0 | series LoRa/GFSK modules (LR1110, LR1120, LR1121) |
| nRF24L01 | 2.4 GHz module |
| RF69 | FSK/OOK radio module |
| RFM2x | series FSK modules (RFM22, RFM23) |
| RFM9x | series LoRa modules (RFM95, RFM96, RFM97, RFM98) |
| Si443x | series FSK modules (Si4430, Si4431, Si4432) |
| STM32WL | integrated microcontroller/LoRa module |
| SX126x | series LoRa modules (SX1261, SX1262, SX1268) |
| SX127x | series LoRa modules (SX1272, SX1273, SX1276, SX1277, SX1278, SX1279) |
| SX128x | series LoRa/GFSK/BLE/FLRC modules (SX1280, SX1281, SX1282) |
| SX123x | FSK/OOK radio modules (SX1231, SX1233) |

Table 3: RadioLib Modules List

## 2.4. Comparison table

| | **Unnoficial Heltec** | **Arduino LoRa** | **RadioLib** |
|---|---|---|---|
| Frequency (MHz) | ☒ | ☒ | ☒ |
| Spreading Factor | ☒ | ☒ | ☒ |
| Signal Bandwidth (KHz) | ☒ | ☒ | ☒ |
| Coding Rate | | ☒ | ☒ |
| Preamble Length (bits) | | ☒ | ☒ |
| $P_{\text{tx}}$ (dBm) | ☒ | ☒ | ☒ |
| Bus Type | SPI | SPI | SPI |
| Callbacks (Channel Activity Detection) | ☒ | ☒ | |
| Implicit / Explicit Header | | ☒ | ☒ |
| Write data to the packet | ☒ | ☒ | ☒ |
| RSSI | | ☒ | ☒ |
| SNR | | ☒ | |
| Packet Frequency Error | | ☒ | |
| Radio modes | | Idle<br>Sleep | |
| Sync word | | ☒ | ☒ |
| CRC | | ☒ | ☒ |
| Invert IQ signals | | ☒ | |
| LNA gain | | ☒ | |

Table 4: Library Table Comparison

# Bibliography

[1] R. Pihlakas, “Heltec ESP32 LoRa V3.” [Online]. Available: https://github.com/ropg/heltec_esp32_lora_v3

[2] J. Gromeš, “RadioLib: Universal wireless communication library for Arduino.” [Online]. Available: https://github.com/jgromes/RadioLib

[3] S. Mistry, “Arduino-LoRa.” [Online]. Available: https://github.com/sandeepmistry/arduino-LoRa

# Acronyms

| Acronym | Meaning |
|---|---|
| LoRa | Long Range |
| MHz | Mega Hertz |
| KHz | Kilo Hertz |
| dBm | decibel-milliwatts |
| SNR | signal-to-noise ratio |
| RSSI | Received Signal Strength Indicator |
| CRC | Cyclic Redundancy Check |
| IQ | In-Phase and Quadrature |
| LNA | Low-Noise Amplifier |